

# HITHA MAGADI VIJAYANAND

✉ hithavgowder@gmail.com ☎ (979) 344-7132 📍 College Station, Texas 🌐 hithamv 🌐 hithamv

## EDUCATION

### Texas A&M University

Master of Computer Science - GPA : 3.77/4.00

Aug 2024 – May 2026  
College Station, Texas

### JSS Science And Technology University

Bachelor of Engineering Computer Science - GPA : 9.52/10.00

Aug 2017 – Jul 2021  
Mysuru, India

## TECHNICAL SKILLS

**Language & Frameworks :** Java, Python, C, C++, JavaScript, Spring Boot, PyTorch, Ruby, Node.js, Typescript, HTML

**Tools & Platforms :** Git, Github, AWS, Kubernetes, Docker, Linux, Unix, Elasticsearch, Terraform, Jenkins, Jira, ServiceNow

**Databases & Others :** MongoDB, SQL, NoSQL, Postgres, DynamoDB, S3, Machine Learning, AI, PyTorch, TensorFlow, SDLC, Agile, REST APIs, Operating Systems, Data Structures and Algorithms, CI/CD Pipelines, DevOps, CloudFormation, Container Services,

**Infrastructure-as-Code (IaC),** Cursor, **Multi-Cloud Support,** Distributed Systems, Cloud-Based Infrastructure

## PROFESSIONAL EXPERIENCE

### JP Morgan Chase

Software Development Engineer 2

Jan 2024 – Jul 2024  
Bengaluru, India

- Migrated microservices from private cloud to AWS **EKS**, improving scalability and **cutting deployment time by 70%**
- Redesigned the user notification workflow using an **event-driven architecture** powered by **RabbitMQ**, enabling the platform to handle and process over 15 million events per month without performance degradation

### JP Morgan Chase

Software Development Engineer 1

Sep 2021 – Dec 2023  
Bengaluru, India

- Led the end-to-end modernization of the platform by upgrading **Spring Boot** from **2.0 to 3.0** and **Java 8** to **Java 17**
- Migrated a legacy MarkLogic system to a cloud-native architecture using AWS S3 and DynamoDB, improving scalability, resolving versioning issues and **reducing infrastructure costs by 80%**
- Engineered AWS OpenSearch for 1500+ users, **improving accuracy by 50%** with synonym-based and context-aware search
- Implemented AWS Elasticsearch indexing for **1M+ records**, engineering custom handlers to boost search speed by 40%
- Created **interactive dashboards** for 5+ teams, streamlining data access and enhancing operational transparency

### JP Morgan Chase

Backend Development Intern

Feb 2021 – Aug 2021  
Bengaluru, India

- Automated business report generation via Outlook and Control-M, ensuring timely delivery and reducing manual effort
- Achieved 100% method coverage with **JUnit** tests and validated performance under load using **JMeter**
- Built scalable **APIs** for creating, updating, and managing Elasticsearch indices, streamlining data indexing workflows and enabling efficient configuration management and system administration

## PROJECTS

### Hybrid Self-RAG - Hallucination Reduction In LLMs | NLP, RAG, OpenAI ☑

Jan 2025 – May 2025

- Built a **Hybrid Self-RAG** system using **Wikipedia** and **ArXiv APIs** improving factual grounding and accuracy by 9%
- Constructed a robust **FactScore-Based** Validation framework utilizing **Sentence-Transformer** embeddings and **cosine similarity** metrics to enhance the factual reliability of LLM outputs, increasing the average FactScore from 0.65 to 0.87

### Generator Matrix Optimization | Evolutionary Algo, Mathematics ☑

Jan 2025 – May 2025

- Implemented a hybrid optimization pipeline combining **linear programming** with **genetic and simulated annealing algorithms** to minimize m-height in generator matrices for linear error-correcting codes
- Enhanced encoding efficiency and fault tolerance for high-reliability communication and storage systems

### Multimodal Network - Advanced Calorie Prediction | CNN, RNN, LSTM ☑

Aug 2024 – Dec 2024

- Designed a multimodal deep learning architecture that integrates **Bi-LSTM** and **CNN** components to fuse CGM time-series data, neural signal features, and meal images, enabling more accurate and context-aware calorie prediction
- Achieved **0.33 RMSRE**, outperforming baselines and validating multimodal integration in health ML models

### Accident Cause Analysis | C++, SQL, Data Mining ☑

Feb 2021 – Jul 2021

- Upgraded the **Association Rule Mining algorithm** to identify accident-prone areas along with **correlated factors**, enabling the traffic department to pinpoint high-risk zones with a 25% improvement in prediction accuracy
- **Published in JETIR (June 2021)**, contributing insights for data-driven traffic safety interventions

## CERTIFICATIONS AND ACHIEVEMENTS

- **Won JP Morgan's national-level hackathon** by developing a scalable solution to transfer large media files to AWS S3 in Aug 2022
- **AWS Cloud Practitioner** certification issued by Amazon Web Services in Jul 2022 - Transferable to Azure